A CRM APPLICATION FOR THECHAISHOP&CO

ABSTRACT

Customer Relationship Management (CRM) systems are essential tools used by businesses to manage customer interactions and improve customer satisfaction. The **CRM Application for TheChaiShop & Co** is designed to help the business store and manage customer data efficiently.

The system provides a centralized platform to maintain customer information, track order history, manage feedback, and monitor sales activities. By digitizing customer data, the application reduces manual work and improves operational efficiency.

This CRM application helps the business understand customer preferences, improve service quality, and build long-term customer relationships. The system includes features such as customer registration, order management, feedback tracking, and sales reporting.

By implementing this CRM system, TheChaiShop & Co can enhance customer engagement, improve decision-making, and support the growth of the business.

1. INTRODUCTION:

Customer Relationship Management (CRM) refers to the strategies, technologies, and practices that organizations use to manage and analyze customer interactions throughout the customer lifecycle. The primary goal of CRM is to improve customer relationships, assist in customer retention, and drive sales growth.

In today's competitive business environment, maintaining strong relationships with customers is very important. Businesses must understand customer needs, track their buying behavior, and provide personalized services.

Small businesses like **TheChaiShop & Co** often face difficulties in managing customer information manually. Traditional methods such as notebooks or spread sheets may lead to data loss, duplication, and inefficiency.

The CRM application is developed to solve these problems by providing a digital system that stores and manages customer information in an organized way. The system allows employees to access customer data quickly and manage orders efficiently.

This application will improve customer satisfaction and help the business make better decisions using customer data.

2. Existing System:

The existing system used in many small businesses involves manual methods of storing customer information and order details. Customer data is usually written in notebooks or stored in basic spread sheets.

This approach has several disadvantages.

First, retrieving customer information becomes difficult and time-consuming. When the number of customers increases, searching through records becomes inefficient.

Second, manual systems are more likely to cause errors such as incorrect entries, duplicate records, and missing information.

Another major issue is the lack of proper order tracking. Businesses may find it difficult to analyze customer purchase history or identify frequent customers.

Customer feedback may also not be recorded properly. As a result, the business cannot analyze customer satisfaction or improve services effectively.

Because of these problems, a more advanced and automated system is required.

3. Proposed System:

The proposed system is a **CRM Application for TheChaiShop & Co** that manages all customer-related activities in a digital platform.

The system provides an easy-to-use interface where staff can store customer information, record orders, and track interactions.

The CRM system allows the business to maintain a complete database of customers. It helps track order history, analyze purchasing patterns, and monitor customer feedback.

The application will also provide administrative features such as login authentication and access control. This ensures that only authorized users can access or modify data.

By implementing the proposed system, the business will benefit from improved data organization, faster access to information, and better customer service.

4. Objectives of the Project:

The main objectives of this CRM application are:

- To develop a system that stores customer information in a centralized database.
- To simplify the process of managing customer data.
- To track customer orders and purchase history.

- To improve communication with customers.
- To collect and manage customer feedback.
- To analyze customer behavior and improve business strategies.
- To increase customer satisfaction and retention.

5. Scope of the Project:

The CRM system will be designed specifically for **TheChaiShop & Co** to manage customer-related activities efficiently.

The scope of the project includes:

- Customer registration and profile management.
- Recording and tracking customer orders.
- Managing customer feedback and complaints.
- Generating reports for sales and customer activity.
- Providing a dashboard for administrators.

The system will mainly be used by shop management and staff members to maintain customer records and improve business operations.

6. System Architecture

The CRM application follows a **three-layer architecture**.

Presentation Layer

The presentation layer represents the user interface of the system. It allows users to interact with the application through forms, dashboards, and reports.

Application Layer

The application layer handles the business logic of the system. It processes user inputs, manages customer data, and performs operations such as adding or updating records.

Database Layer

The database layer stores all the data related to customers, orders, and feedback. The database ensures that information is securely stored and easily accessible.

7. Modules of the System:

The CRM application consists of several modules.

1. User Management Module

This module manages user authentication and access control. Only authorized staff and administrators can access the system.

2. Customer Management Module

This module stores customer details such as name, phone number, email, and address.

3. Order Management Module

This module records and tracks customer orders. It maintains the order history of each customer.

4. Feedback Management Module

This module collects and manages customer feedback and complaints.

5. Reporting Module

This module generates reports related to sales, customer activities, and feedback analysis.

8. System Requirements:

Hardware Requirements

- Computer or Laptop
- Minimum 4GB RAM
- 500GB Hard Disk

- Internet connection

Software Requirements

- Operating System: Windows / Linux
- Programming Language: Java / Python / PHP
- Database: MySQL
- Frontend: HTML, CSS, JavaScript
- Development Tools: Visual Studio Code

9. Database Design:

The CRM system uses a database to store and manage information.

Customers Table

Fields include:

- Customer_ID
- Name
- Phone_Number
- Email
- Address
- Registration_Date

Orders Table

Fields include:

- Order_ID
- Customer_ID
- Product_Name
- Quantity
- Price
- Order_Date

Feedback Table

Fields include:

- Feedback_ID
- Customer_ID
- Comments
- Rating
- Date

10. Advantages of the System:

The CRM application provides several advantages:

- Centralized data management
- Easy access to customer information
- Improved customer service
- Better sales tracking
- Reduced manual work
- Improved decision making

11. Limitations:

Although the system provides many benefits, it also has some limitations.

- Initial setup and configuration may require time.
- Staff may need training to use the system effectively.
- Data security must be maintained carefully.

12. Future Enhancements:

The system can be improved with additional features in the future.

Possible enhancements include:

- Mobile application for easier access
- Integration with online payment systems
- Customer loyalty programs
- Automated SMS and email notifications
- Advanced analytics and reporting tools

13. Conclusion:

The **CRM Application for TheChaiShop & Co** provides an efficient solution for managing customer relationships and business operations. The system helps store customer data, track orders, and manage feedback in an organized manner.

By implementing this CRM system, the business can improve customer satisfaction, increase operational efficiency, and support long-term growth.

The application demonstrates how digital tools can help small businesses manage customer relationships effectively.

14. Data Flow Diagram (DFD):

A **Data Flow Diagram (DFD)** represents how data moves through the system and how it is processed. It shows the interaction between users, processes, and databases.

The DFD helps in understanding how information flows within the CRM application and how different modules communicate with each other.

In the **CRM Application for TheChaiShop & Co**, the DFD illustrates how customer data, orders, and feedback are processed and stored in the system.

The system mainly involves three components:

- External entities (Admin and Staff)
- Processes (Customer Management, Order Management, Feedback Management)
- Data stores (Customer Database, Orders Database, Feedback Database)

14.1 DFD Level 0 (Context Diagram):

The **Level 0 DFD** represents the entire system as a single process. It shows how external users interact with the CRM system.

External Entities

- Admin

- Staff
- Customer

Main Process

CRM System

Data Flow

Admin / Staff can:

- Add customer details
- Update order information
- View reports

Customer can:

- Provide feedback
- Place orders

The CRM system stores all this data in the database and provides reports to the admin.

How to draw in Word

Insert → Shapes

Draw:

Customer → CRM System → Database
Admin → CRM System → Reports

Label arrows as:

- Customer Data
- Order Details
- Feedback
- Reports

14.2 DFD Level 1:

The **Level 1 DFD** breaks the CRM system into smaller processes.

Processes

1. Customer Management
2. Order Management
3. Feedback Management
4. Report Generation

Data Stores

- Customer Database
- Order Database
- Feedback Database

Flow

Admin enters customer details → Customer Management → Customer Database

Customer places order → Order Management → Order Database

Customer gives feedback → Feedback Management → Feedback Database

Admin requests report → Report Generation → Display Report

15. Entity Relationship Diagram (ER Diagram):

The **Entity Relationship (ER) Diagram** shows the relationship between different entities in the database.

Entities in the CRM System

1. Customer
2. Order
3. Feedback
4. Admin

Attributes

Customer

- Customer_ID (Primary Key)
- Name
- Phone
- Email
- Address

Order

- Order_ID (Primary Key)
- Customer_ID (Foreign Key)
- Product_Name
- Quantity
- Price
- Order_Date

Feedback

- Feedback_ID
- Customer_ID
- Comments
- Rating

Admin

- Admin_ID
- Username
- Password

Relationships

Customer → Places → Order
Customer → Gives → Feedback
Admin → Manages → System

How to draw

Use rectangles for entities, ovals for attributes, diamonds for relationships.

Example:

Customer — Places — Order
Customer — Gives — Feedback

16. Use Case Diagram:

A **Use Case Diagram** describes how users interact with the system.

Actors

- Admin
- Staff
- Customer

Use Cases

Admin can:

- Login
- Add Customer
- Update Customer
- Manage Orders
- View Reports
- Manage Feedback

Customer can:

- Place Order
- Give Feedback

Diagram Structure

Admin → Login
Admin → Manage Customers
Admin → View Reports

Customer → Place Order
Customer → Submit Feedback

17. System Flow Diagram:

The **System Flow Diagram** explains the step-by-step working process of the CRM system.

Steps

1. User opens the CRM application.
2. Admin or staff logs into the system.
3. After login, the dashboard is displayed.
4. User can choose different modules such as:
    - Customer Management
    - Order Management
    - Feedback Management
5. Data entered by the user is stored in the database.
6. Reports can be generated based on stored data.
7. User logs out after completing tasks.

This flow ensures smooth data management and efficient system operation.

18. Testing:

Testing is performed to ensure that the CRM application works correctly and meets the requirements.

Types of Testing

Unit Testing
Each module such as customer management and order management is tested individually.

Integration Testing
Different modules are tested together to ensure proper communication.

System Testing
The entire system is tested to verify that all functionalities work as expected.

User Acceptance Testing
The system is tested by users to ensure it meets their needs.

19. Feasibility Study:

A **feasibility study** is conducted to determine whether the proposed system is practical and beneficial for the organization. It evaluates different aspects such as technical feasibility, economic feasibility, and operational feasibility.

Technical Feasibility

Technical feasibility examines whether the required technology and resources are available to develop the CRM system. The proposed CRM application can be developed using common technologies such as HTML, CSS, JavaScript, and MySQL database. These technologies are widely used and easily available.

The system can run on standard computers with basic hardware requirements. Therefore, the project is technically feasible.

Economic Feasibility

Economic feasibility evaluates the cost involved in developing and maintaining the system. Since the CRM application can be developed using open-source tools and software, the cost of development is relatively low.

The benefits of the system, such as improved customer management and increased efficiency, outweigh the development cost. Hence, the project is economically feasible.

Operational Feasibility

Operational feasibility checks whether the system will be accepted and used by employees. The CRM application is designed with a user-friendly interface so that staff members can easily learn and use the system.

The system simplifies daily operations and reduces manual work, which makes it highly beneficial for the organization.

20. Security and Privacy:

Security is an important aspect of any software system. The CRM application must ensure that customer data is protected from unauthorized access.

The system includes **user authentication**, where users must log in with a username and password to access the application. This prevents unauthorized users from accessing sensitive information.

Data privacy is also maintained by restricting access to certain modules based on user roles. For example, only administrators can modify or delete records.

Additionally, regular data backups can be implemented to prevent data loss.

21. Benefits to the Organization:

The CRM application provides several benefits to **TheChaiShop & Co**.

First, it helps maintain a well-organized database of customers, which improves customer relationship management.

Second, it allows the business to track customer purchase history and identify frequent customers.

Third, it helps analyze customer feedback, which allows the business to improve its products and services.

The system also saves time and reduces manual work, enabling staff members to focus on providing better service to customers.

22. Impact of CRM on Business Growth:

Implementing a CRM system can have a positive impact on business growth.

By maintaining detailed customer records, the business can understand customer preferences and provide personalized services. This leads to higher customer satisfaction.

CRM systems also help businesses identify sales trends and customer behavior patterns. These insights can help the organization develop better marketing strategies and increase sales.

For **TheChaiShop & Co**, the CRM system can help build stronger customer relationships and improve long-term profitability.

23. Summary:

The CRM Application for TheChaiShop & Co is developed to improve customer relationship management by providing a centralized system for managing customer information.

The system includes various modules such as customer management, order tracking, feedback management, and reporting. These modules work together to ensure efficient data management and improved customer service.

By using this system, the business can streamline its operations, reduce manual work, and enhance decision-making.

The project demonstrates how digital solutions can help small businesses improve efficiency and achieve sustainable growth.

24.Importance of CRM Systems:

Customer Relationship Management systems have become an essential tool for modern businesses. They help organizations manage customer interactions in a structured and efficient manner.

A CRM system allows businesses to collect, store, and analyze customer information in a centralized database. This information can include customer contact details, purchase history, feedback, and communication records.

For businesses like **TheChaiShop & Co**, a CRM system helps improve customer service by allowing staff members to quickly access customer information and understand their preferences.

CRM systems also help businesses build stronger relationships with customers by providing personalized services and timely communication. This results in higher customer satisfaction and loyalty.

25. Role of CRM in Small Businesses:

Small businesses often face challenges in managing customer information due to limited resources and manual processes. A CRM system can help small businesses organize their customer data and improve business operations.

For example, a tea shop like **TheChaiShop & Co** may have many regular customers. Without a proper system, it becomes difficult to track their orders, preferences, or feedback.

By implementing a CRM system, the shop can easily maintain records of customer purchases, analyze sales patterns, and identify loyal customers.

This helps the business provide better services and develop marketing strategies that attract more customers.

26.Customer Data Management:

Customer data management is one of the most important features of a CRM system. It involves collecting, storing, and organizing customer information in a structured format.

The CRM application for **TheChaiShop & Co** maintains a customer database that contains details such as:

- Customer name
- Contact number
- Email address
- Address
- Purchase history
- Feedback and ratings

This information allows the business to understand customer behavior and improve services.

Proper data management also helps prevent data duplication and ensures that customer information remains accurate and up to date.

27. Order Tracking System:

The order tracking system is another important module of the CRM application. It helps record and monitor all customer orders.

Each order placed by a customer is stored in the database along with important details such as:

- Order ID
- Customer ID
- Product name
- Quantity
- Price
- Order date

This information helps the business track sales and analyze customer buying patterns.

Order tracking also helps the management identify popular products and manage inventory more effectively.

28. Customer Feedback Analysis:

Customer feedback is an important source of information that helps businesses improve their services.

The CRM system includes a feedback management module where customers can provide comments, suggestions, or ratings.

This feedback is stored in the database and can be analyzed by the management to identify areas that require improvement.

For example, if several customers complain about a particular issue, the business can take immediate action to resolve it.

By analyzing customer feedback, **TheChaiShop & Co** can enhance customer satisfaction and build stronger relationships with its customers.

29. Reporting and Analytics:

The CRM system provides reporting and analytics features that help management make informed business decisions.

Reports can be generated based on different criteria such as:

- Customer activity
- Sales performance
- Order history
- Customer feedback

These reports provide valuable insights into business operations and help identify trends and patterns.

For example, the system can generate reports showing which products are most popular among customers.

Such information helps the business improve its product offerings and marketing strategies.

30. Data Backup and Recovery:

Data backup and recovery are essential for maintaining the reliability of the CRM system.

Regular backups ensure that important customer information is not lost in case of system failure or technical issues.

The system can store backup copies of the database at regular intervals. If data loss occurs, the backup can be used to restore the system to its previous state.

This ensures that business operations are not disrupted and that customer data remains secure.

31. User Training:

After the CRM system is developed and implemented, employees must be trained to use it effectively.

Training helps staff members understand the different features and modules of the system.

The training process may include:

- Introduction to the CRM system
- Instructions on how to add or update customer records
- Managing orders and feedback
- Generating reports

Proper training ensures that the system is used efficiently and helps the organization achieve its goals.

32. Implementation Strategy

The implementation strategy describes how the CRM system will be introduced and used in the organization.

The implementation process typically includes the following steps:

1. System installation
2. Database setup
3. User account creation
4. Staff training
5. Testing and verification
6. Full system deployment

Once the system is fully implemented, it becomes part of the daily operations of the business.

33. Project Outcome:

The successful implementation of the CRM application provides several positive outcomes for **TheChaiShop & Co**.

The system improves the way customer information is stored and managed. It also helps the business track orders, analyze feedback, and generate useful reports.

As a result, the business can provide better customer service and improve overall operational efficiency.

The CRM application demonstrates how technology can help small businesses improve their customer relationship management processes.

34.References:

The following resources were used during the development of this project.

- Software Engineering Concepts
- Database Management System Books
- Online documentation and tutorials
- Research articles related to CRM systems

Websites:

- www.w3schools.com
- www.geeksforgeeks.org
- www.mysql.com